

Modelica, in turn, finds use in a variety of applications like fluid systems (steam power generation, hydraulics), automotive (power-trains, transmission systems) and mechanical systems (dual-body systems, mechatronics). It is also being increasingly used in the simulation process of thermo-fluid and energy systems.

OpenModelica and its components

The goal envisioned by the creators and managers of OpenModelica was to "... create a comprehensive open source Modelica modeling, compilation and simulation environment based on free software distributed in binary and source code form for research, teaching and industrial usage." Now, OpenModelica is not the most-commonly-used variant of Modelica ecosystem but a lot of industrial products use it along with proprietary software to simulate applications in the field of power plant optimisation, automotive and water treatment, to name a few.

OpenModelica Compiler(OMC). This is the compiler used by OpenModelica to convert objects from the base Modelica code to a C code. The user can use APIs available freely to interact with the system through Advanced Interactive OpenModelica Compiler. This is the crucial part of the package as it acts like the engine of the simulator.

OpenModelica Connection Editor (OMEdit). This is a graphical user interface that lets users create and modify models on OpenModelica. It uses OMC by means of a C language API to perform the requisite simulation, model query and plotting.

OpenModelica Shell (OMShell).

This is an interactive command-line interface that handles sessions and does simple editing work like parsing. This is where you type your commands, much like DOS screen command-line interface.

OpenModelica Notebook (OMNotebook). Developed more as an advanced teaching tool, OMNotebook is a lightweight mathematical editor for Modelica. If you have used the niche search engine WolframAlpha, then you would know what this is.

Short- and long-term goals

The community driving OpenModelica aims to develop a complete implementation along with an efficient, interactive computational environment for Modelica language. In the long run, however, the team hopes to be able to build a complete reference implementation of Modelica language, including simulation of equation based models, additional facilities in the programming environment as well as research in new activities.

Another area to work on is language design so as to improve

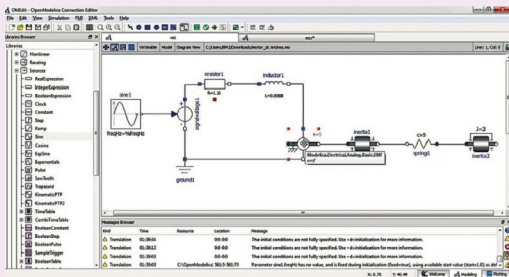


Fig. 2: OMEdit

OpenModelica Python Interface (OMPython). This component contains a Python based package and API for accessing the functionalities made available by OMC.

Modelica Development Tooling (MDT). This package is also an advanced tool for model editing and uses code assistance to solve its problems. MDT communicates with OMC through an API that is based on Common Object Request Broker Architecture (CORBA), a standard developed to facilitate communication of systems that are deployed on diverse platforms.

On March 16, 2016, OpenModelica 1.9.6 was released with updates to fix an FMU generation bug.

properties such as orthogonality, declarativity, reuse, configurability and so on.

Online help and documentation on OpenModelica is extensive and easy to find. Because of its use in academia, you can be rest assured that all FAQs are explained in detail, and you can find examples of code and a lot of relevant detail at openmodelica.org. If simulation feels like something you want to explore, you should definitely install the latest version (available in this month's DVD) and give it a whirl! **EFY**

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